6 A ball is thrown horizontally with a speed of $10.0\,m\,s^{-1}$ above horizontal ground. The ball hits the ground after a time of 3.0 s.

Air resistance is negligible.

What is the speed of the ball just before it hits the ground?

A $10\,m\,s^{-1}$ **B** $29\,m\,s^{-1}$ **C** $31\,m\,s^{-1}$ **D** $39\,m\,s^{-1}$